

A Crash Course in Arduino Programming

Lesson 1 Exercises

1. Find an Arduino sketch you have written or one that has been given to you as an example. What are some of the variables used in the sketch? What was the purpose of those variables? Do you think the author chose appropriate datatypes?
2. Try to produce an error message in the Arduino IDE that we did not cover in this lecture. Explain how you produced the error and the error message the IDE gave you. If you didn't already know what caused the error, how could you use the given error message to guide your troubleshooting process?

3. Find the official Arduino documentation website. Find a datatype that is not described in the lesson handout and explain its properties here. What would be an appropriate use case for that datatype?

4. What is an application for which an Arduino would be better suited than a Raspberry Pi? What is an application for which a Raspberry Pi would be better suited than an Arduino? Can you think of a situation in which it would be useful to use both an Arduino and a Raspberry Pi together?

5. Assuming an appropriately functioning Arduino and program, what is the value of myNum after each of the following operations?

```
int myNum = 15%6;  
myNum += 4;  
myNum--;
```

6. Find the errors in the following code and propose a functioning solution.

```
void setup():  
  Serial.println("Hello World!");  
  int aDog = "Golden Retriever";  
  
void loop() {  
  Serial.println(aDog)  
}
```

7. You have likely worked primarily with the Arduino Uno in the past. Find the documentation for any other Arduino board and explain what it has in common with and how it differs from the Uno.